

November 7: The Randy Hall Problem

Puzzle

You are a producer on a game show hosted by Randy "Random" Hall (no relation to Monty Hall). The show has three doors labeled 1 through 3 from left to right, and behind them are various prizes.

Contestants pick one of the three doors at which to start, and then they press an electronic button many, many times in rapid succession. Each time they press the button, they either stay at their current door or move to an adjacent door. If they're at door 2 and move to an adjacent door, that new door will be 1 or 3 with equal probability.

Randy has decided that when a contestant presses the button while at door 2, there should be a 20 percent chance they remain at door 2.

As the producer, you want the chances of a contestant ultimately winding up at each of the three doors to be nearly equal after many button presses. Otherwise, mathematicians will no doubt write you nasty letters complaining about how your show is rigged.

If a contestant presses the button while at door 1 (or door 3), what should the probability be that they remain at that door?

Solution

This looks like a Markov chain problem yay! Let p be the probability of staying at door 1 (or door 3) after hitting the button. Then we can model this a Markov chain with three states and transition probability matrix below:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} p & 1-p & 0 \\ 0.4 & 0.2 & 0.4 \\ 0 & 1-p & p \end{bmatrix} \end{matrix}$$

We are told that in the steady state, landing on each door should be equally likely after hitting the button many times. This means that the stationary distribution will be described by:

$$\vec{\pi} = [\pi_1 \quad \pi_2 \quad \pi_3] = [\pi \quad \pi \quad \pi] = \left[\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}\right]$$

Faithful to its name, the stationary distribution will be unchanged after an operation with the transition probability matrix: $\vec{\pi}P = \vec{\pi}$, and so we have:

$$[\pi \quad \pi \quad \pi] \begin{bmatrix} p & 1-p & 0 \\ 0.4 & 0.2 & 0.4 \\ 0 & 1-p & p \end{bmatrix} = [\pi \quad \pi \quad \pi]$$

The first equation we get when we carry out this product is:

$$\pi p + 0.4\pi = \pi$$

$$p + 0.4 = 1$$

$$\boxed{p = 0.6}$$